

GUJARAT TECHNOLOGICAL UNIVERSITY**V.V.P. Engineering College, Rajkot**

Department of Computer Engineering

PROJECT BASED LEARNING (PBL) REPORT**Computer Network (BE05000171)**

PBL Activity	Title
Activity No. 2 (Industry Visit)	Visit to Data Centre – Rajkot Nagarik Sahakari Bank (RNSB) – Industrial Visit Report (10 hours)
Activity No. 8 (Case Study)	Case study on network outage – Facebook/Meta BGP & DNS outage (2021) and AWS us-east-1 DNS outage (2025) (5 hours)

1. Student Details

Student Name	Hetshree Trivedi
Enrollment No. / Roll No.	240470107060 / 24ce181
Semester & Division	5th Semester – CE, Batch GX-4
Branch / Department	Computer Engineering (CE)
Academic Year	2026–27

Contents: Part A – Industrial Visit Report (Sections 2–10) | Part B – Case Study on Network Outage (Section 11 onwards)

PART A – INDUSTRIAL VISIT REPORT (Activity 2)

2. Industrial Visit Details

Name of Industry / Company	RNSB (Rajkot Nagarik Sahakari Bank Ltd.) Data Centre
Industry Address	Dhebar Road, Opp. Bank of Baroda, Para Bazar Chowk, Rajkot, Gujarat 360001
Date of Visit	12/08/2026 (Batch 1) and 13/08/2026 (Batch 2)
Faculty Coordinator	Prof. Ankur Kaneriya
Faculty Involved – 12/08/2026	Prof. Ankur Kaneriya and Prof. Vidhi Rajani
Faculty Involved – 13/08/2026	Savan Vachhani and Prof. Pooja Oza
Industry Persons	Mr. Gautam Renuka, Mr. Ravi Jani

3. Introduction

About the organisation. Rajkot Nagarik Sahakari Bank Ltd. (RNSB) is a leading co-operative bank headquartered in Rajkot, Gujarat. It is a Multi-State Scheduled Co-operative Bank operating in Gujarat and Maharashtra.

Establishment and background. The bank was established on 5 October 1953 with a small capital of Rs. 4,890 contributed by 59 members, and was the first co-operative institution to start functioning in the erstwhile state of Saurashtra. The bank began automation and modernisation in 1987 and had computerised all of its branches by 1995. It has held Scheduled Bank status since 1988 and was registered under the Multi-State Co-operative Societies Act in 2001. According to the bank's published figures (31 March 2021), it had 38 branches and more than 3 lakh members.

Main products / services. Savings and current accounts, fixed and recurring deposits, retail and business loans, NEFT and other electronic fund transfers, mobile banking and internet banking, ATM services, and depository-participant (CDSL) services. All of these depend on a reliable central IT infrastructure – the data centre we visited.

Industry sector. Banking and financial services (co-operative banking), supported by an in-house IT / data-centre operation.

4. Objectives of the Visit

Purpose. To see how a real bank runs its data centre and to connect the Computer Network syllabus (LAN, WAN, switching, routing, security, redundancy) with a live, business-critical installation.

Learning objectives

- Understand the layout and components of a production data centre.

- Observe how branches, ATMs and mobile/internet channels connect to the core banking system.
- Learn how availability, backup, disaster recovery and security are achieved in practice.
- Relate the OSI/TCP-IP layers, VLANs, firewalls and routing concepts to real equipment.
- Understand career skills required in IT infrastructure and network administration.

Technical concepts we expected to understand

- Network topology, structured cabling, switching and routing in a data centre.
- Redundancy and high availability (dual links, dual power, clustering).
- Network security: firewall, DMZ, access control, monitoring.
- Servers, storage, database, backup and disaster recovery (DR) site.

5. Industry Profile

Organisation structure. A typical structure of a multi-state co-operative bank, as explained to us, is shown below.

Level	Role
Board of Directors	Elected by members; policy making and oversight.
Chief Executive / Management	Day-to-day administration, regulatory compliance with RBI.
Departments	Operations, Credit / Advances, Accounts, Audit & Compliance, HR, Information Technology.
Branch network	Branches and extension counters delivering services to customers.

Major departments. Information Technology (data centre, network, security, application support), Operations, Credit, Audit and Compliance, Customer Service.

Technologies used. Core Banking System (CBS) and database servers, ATM / payment switch, internet and mobile banking platforms, firewalls and network security devices, WAN links to branches, backup and storage systems, UPS and diesel generator for power backup.

Major clients / operations. Retail depositors, small businesses, salaried customers and co-operative members across Gujarat and Maharashtra, served through branches, ATMs and digital channels.

6. Visit / Technical Learning

6.1 Activities observed

1. Reception at the RNSB building on Dhebar Road and an introduction to the bank and its IT operations by Mr. Gautam Renuka and Mr. Ravi Jani.
2. Guided walk-through of the data centre / server room, where the industry person explained the racks and their contents (see photographs in Section 9).

3. Explanation of how branch traffic reaches the core banking servers and how services stay available.
4. Question–answer session with the technical staff (Section 8).

6.2 Machines, equipment and software observed

- **Enclosed server and network racks** with transparent glass doors and blue LED lighting, holding servers, switches and patch panels.
- **Structured cabling** – neatly dressed copper/fibre cables with labelled patch panels inside the racks.
- **Raised-floor tiles and false ceiling** – used for airflow, cable routing and housing of detectors/lighting.
- **Controlled-entry server room** with restricted access; visitors entered in groups with staff supervision.
- **Diesel generator (DG)** set parked beside the building entrance, acting as backup power when mains supply fails.
- **Software / services:** core banking application, database, monitoring tools and backup software (explained verbally).

6.3 Networking and IT infrastructure

A bank network is built in layers. Branch LANs (switches and PCs) connect through branch routers to a WAN (leased line / MPLS / VPN). At the data centre the WAN terminates on edge routers, passes through a pair of firewalls and reaches the core switches, which connect the application, database, ATM/UIP and storage servers. Internet-facing services such as internet banking sit in a separate DMZ, and data is replicated to a Disaster Recovery (DR) site. Figure 1 shows this design; I drew it in the style of a Cisco Packet Tracer topology for clarity.

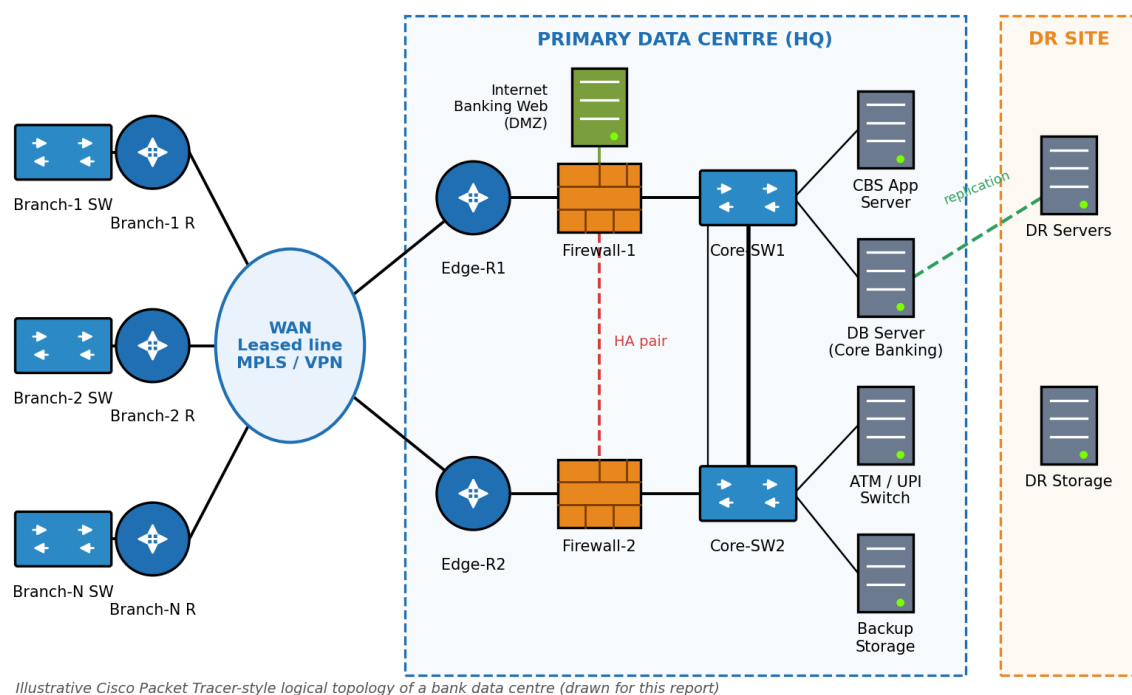
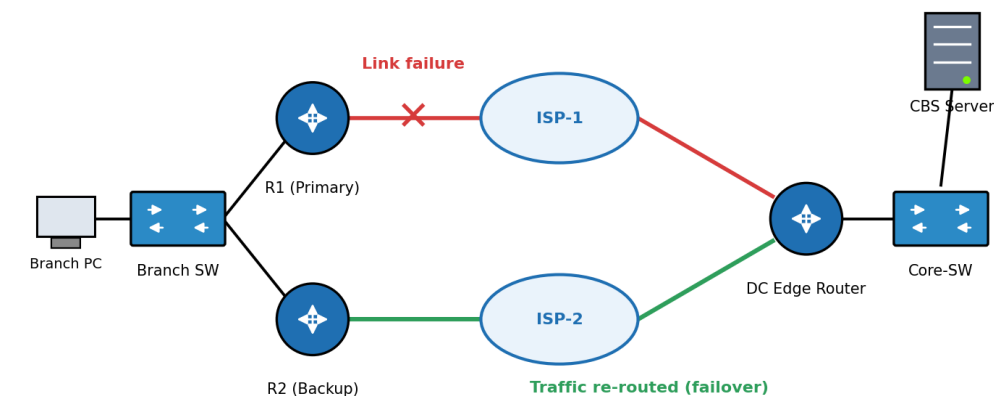


Figure 1: Packet Tracer-style logical topology of a bank data centre (illustrative)

Redundancy. Every critical element is duplicated – two WAN links, two edge routers, two firewalls in a high-availability pair and two core switches. Gateway redundancy is usually achieved with a First-Hop Redundancy Protocol such as HSRP. A configuration I practised on Packet Tracer for the core switches is shown below.

```
Core-SW1(config)# interface vlan 10
Core-SW1(config-if)# ip address 10.10.10.2 255.255.255.0
Core-SW1(config-if)# standby 10 ip 10.10.10.1      ! virtual gateway
Core-SW1(config-if)# standby 10 priority 110        ! active router
Core-SW1(config-if)# standby 10 preempt

Core-SW2(config-if)# standby 10 ip 10.10.10.1      ! standby, default priority 100
```



Illustrative Packet Tracer-style scenario: redundant WAN links keep the service alive when one path fails

Figure 2: Packet Tracer-style failover – traffic moves to the backup WAN link when the primary fails (illustrative)

6.4 Security, database, servers and power

Area	What we learned
Physical security	Restricted access to the server room, supervised entry, CCTV and access logging.
Network security	Firewalls between WAN, internet and internal zones; DMZ for internet banking; network segmentation using VLANs; monitoring of traffic and logs.
Servers & database	Application and database servers run the core banking system; clustering and regular maintenance keep it available.
Backup & DR	Regular data backups and replication to a DR site so operations can resume after a major failure; DR drills are performed periodically.
Power & cooling	UPS for short interruptions, diesel generator for long outages, and air-conditioning to keep equipment within temperature limits.
Compliance	Banking IT follows RBI guidelines on IT governance, cyber security and business continuity.

Use of Wireshark. Network engineers analyse traffic with a packet analyser such as Wireshark to diagnose problems. Useful display filters I practised: `dns`, `bgp`, `icmp.type == 3` and `tcp.analysis.retransmission`. Sample views are in Part B.

7. Student Learning Outcomes

What I learned. A data centre is much more than servers: power, cooling, cabling, security and network design together decide whether a bank can serve customers 24×7.

- **New technologies / concepts:** DMZ design, HA firewall pairs, HSRP, DR replication, structured cabling, raised-floor data-centre design.
- **Skills gained:** reading a network topology, relating device roles to OSI layers, basic Packet Tracer configuration, packet analysis with Wireshark, technical report writing.
- **Industry practices:** change control, redundancy for every single point of failure, periodic DR drills, strict access control and regulatory compliance.

Difference between theoretical and practical knowledge

Aspect	Classroom (theory)	Data centre (practice)
Topology	Simple diagrams with a few devices	Dual everything, cable management, labelling, strict layout
Reliability	Concept of fault tolerance	UPS + DG + dual links + DR site tested regularly
Security	Firewall/ACL definitions	Layered physical and network security, monitoring, audits
Failures	Ideal conditions	Planned maintenance windows, rollback plans, incident handling

8. Interaction & Q&A

Students interacted with Mr. Gautam Renuka, Mr. Ravi Jani and the technical staff during and after the walk-through. The main questions and the explanations given are summarised below.

Question asked by students	Answer / explanation by industry experts
How is 24×7 availability of banking services ensured?	By removing single points of failure: redundant links, devices and servers, plus UPS and a diesel generator for power backup.
How are branches connected to the data centre?	Through dedicated WAN links (leased line / MPLS / VPN) with backup links; traffic is protected and monitored at the data centre edge.
What is the difference between the data centre and the DR site?	The data centre runs day-to-day operations; the DR site holds replicated data and systems so the bank can continue if the primary site fails.
How is the network protected from cyber-attacks?	Firewalls, network segmentation, restricted access, regular patching, log monitoring and periodic audits as per RBI guidelines.
How often are backups taken and tested?	Backups are taken regularly and restored/drilled periodically to be sure that data can actually be recovered.

Question asked by students	Answer / explanation by industry experts
Which skills should we build for a career in IT infrastructure?	Networking fundamentals (CCNA level), Linux/Windows servers, databases, security basics and, increasingly, cloud technologies.

9. Photographs

9.1 Group photographs



Batch 1 with faculty and industry staff outside RNSB – 12/08/2026



Batch 2 with faculty and industry staff inside RNSB – 13/08/2026

9.2 Industry / company photograph



RNSB building entrance on Dhebar Road with the backup diesel generator on the left – 12/08/2026

9.3 Technical demonstration photograph



Industry person explaining the server and network racks in the data centre – 12/08/2026

9.4 Student interaction photograph



Students interacting with the industry person during the data-centre walk-through – 13/08/2026

10. Conclusion

Overall experience. The visit to the RNSB data centre was a valuable, practical extension of the Computer Network course. Seeing real racks, cabling, power backup and security arrangements made abstract topics such as redundancy, segmentation and disaster recovery easy to understand.

Key takeaways. (1) High availability comes from duplicating every critical component; (2) security has both physical and network layers; (3) backup and DR must be tested, not just configured; (4) the network must be documented and managed with discipline.

Usefulness. The visit improved my understanding of how enterprise networks are designed and operated, motivated me to practise with Cisco Packet Tracer and Wireshark, and gave me a clear picture of skills required for a career in networking and IT infrastructure. I thank the faculty coordinators, Mr. Gautam Renuka, Mr. Ravi Jani and the RNSB management for this opportunity.

PART B – CASE STUDY ON NETWORK OUTAGE (Activity 8)

11. Introduction

A network outage is an unplanned loss of connectivity or of the services that depend on it. For a bank, an airline or a social network, even minutes of downtime mean lost money and lost trust. This case study examines two major outages – one caused by a routing (BGP) and DNS failure at Facebook in 2021 and one caused by a DNS automation defect at Amazon Web Services in 2025 – and extracts lessons that apply to any network, including the bank data centre we visited.

12. Case 1: Facebook (Meta) outage – 4 October 2021

What happened. On 4 October 2021, Facebook, Instagram, WhatsApp and Messenger became unreachable worldwide for roughly six hours. Users saw errors because the name “facebook.com” could no longer be resolved to an IP address.

Root cause. According to Meta's engineering report, during routine maintenance a command intended to assess the availability of global backbone capacity unintentionally took down all connections in the backbone network, disconnecting Facebook's data centres from each other and from the internet. A bug in the audit tool that should have blocked such a command failed to stop it.

Why DNS and BGP made it worse. Facebook's authoritative DNS servers are designed to withdraw their BGP route advertisements if they cannot reach the data centres, because that indicates an unhealthy network. When the backbone went down, every DNS location did exactly that. The BGP prefixes that contained the DNS servers disappeared from the internet's routing tables, so the DNS servers became unreachable even though they were still running. Figure 3 shows what a Wireshark capture of such an event looks like.

Capturing from Ethernet1 - BGP session with upstream peer					
Filter: bgp dns icmp					
No.	Time	Source	Destination	Protocol	Info
1	0.000	203.0.113.1	203.0.113.2	BGP	KEEPALIVE Message
2	12.4	203.0.113.1	203.0.113.2	BGP	UPDATE Message, Withdrawn Routes: 198.51.100.0/24
3	12.4	203.0.113.1	203.0.113.2	BGP	UPDATE Message, Withdrawn Routes: 192.0.2.0/24
4	12.5	203.0.113.1	203.0.113.2	BGP	UPDATE Message, Withdrawn Routes: 203.0.113.0/24
5	15.0	192.0.2.10	198.51.100.53	DNS	Standard query 0x3c4d A www.example-social.com
6	20.0	192.0.2.10	198.51.100.53	DNS	Standard query 0x3c4d A www.example-social.com [No response]
7	25.1	192.0.2.10	192.0.2.1	ICMP	Destination unreachable (Network unreachable)

> Border Gateway Protocol - UPDATE Message

Type: UPDATE Message (2)

Withdrawn Routes Length: 4

Withdrawn Routes: 198.51.100.0/24 <-- prefix holding the DNS name servers

[Path attributes: none -> route removed from the internet routing table]

Illustrative Wireshark-style view built for this report (documentation IP ranges); not a live capture.

Figure 3: Wireshark-style view – BGP UPDATE messages withdrawing the prefix of the DNS servers, followed by failing DNS queries (illustrative)

Impact and recovery. Internal tools that depended on the same network and DNS stopped working, and even remote (out-of-band) access was down. Engineers had to be sent physically to the data centres, where strong physical and system security slowed the recovery. After the

backbone was restored, services were brought back gradually to avoid overloading the power and systems.

13. Case 2: AWS US-EAST-1 (DynamoDB DNS) outage – 19/20 October 2025

What happened. Between 11:48 PM PDT on 19 October and 2:20 PM PDT on 20 October 2025, Amazon Web Services suffered a major disruption in its Northern Virginia (us-east-1) Region. Many popular applications and websites that depend on AWS were affected.

Root cause. AWS's post-event summary traces the incident to a latent race condition in the automated DNS management system of Amazon DynamoDB. It produced an incorrect empty DNS record for the regional endpoint dynamodb.us-east-1.amazonaws.com, and the automation failed to repair it. All systems that needed to reach DynamoDB through the public endpoint got DNS failures, including AWS's own internal services.

Cascading effect. Because many AWS services depend on DynamoDB, the failure spread: new EC2 instance launches failed for several hours and Network Load Balancers saw connection errors due to failing health checks. Some customers reported problems for up to about 15 hours. Figure 4 shows how such a DNS failure looks in Wireshark.

Capturing from Ethernet0 - client view of the DNS failure					
Filter: dns.qry.name contains "dynamodb"					
No.	Time	Source	Destination	Protocol	Info
1	0.000	192.0.2.10	198.51.100.53	DNS	Standard query 0x1a2b A dynamodb.us-east-1.amazonaws.com
2	0.021	198.51.100.53	192.0.2.10	DNS	Standard query response 0x1a2b A (no answer records)
3	1.004	192.0.2.10	198.51.100.53	DNS	Standard query 0x1a2c A dynamodb.us-east-1.amazonaws.com
4	1.026	198.51.100.53	192.0.2.10	DNS	Standard query response 0x1a2c A (no answer records)
5	3.010	192.0.2.10	198.51.100.53	DNS	Standard query 0x1a2d A dynamodb.us-east-1.amazonaws.com
6	3.031	198.51.100.53	192.0.2.10	DNS	Standard query response 0x1a2d A (no answer records)
7	5.200	192.0.2.10	-	TCP	[Client cannot open connection: no IP address to connect to]
> Domain Name System (response) Flags: 0x8180 Standard query response, No error Questions: 1 Answer RRs: 0 Authority RRs: 0 Queries: dynamodb.us-east-1.amazonaws.com: type A, class IN [Empty answer section -> client has no IP address to connect to]					

Illustrative Wireshark-style view built for this report (documentation IP ranges); not a live capture.

Figure 4: Wireshark-style view – DNS responses with no answer records, so the client cannot connect (illustrative)

14. Comparison of the two outages

Aspect	Facebook (2021)	AWS (2025)
Trigger	Faulty maintenance command on the backbone; audit-tool bug	Latent race condition in DNS automation
Layer involved	Network layer routing (BGP) and application layer DNS	Application layer DNS, then cloud control plane
Symptom	DNS servers unreachable; BGP prefixes withdrawn	Empty DNS record; endpoint could not be resolved
Duration	About 6 hours	About 14.5 hours (23:48 to 14:20 PDT)

Aspect	Facebook (2021)	AWS (2025)
Recovery challenge	No remote access; physical access to data centres needed	Automation could not self-repair; manual fix and gradual recovery
Common theme	A small configuration or automation error spread across a tightly coupled system	Same

15. Analysis and lessons learned

1. **Avoid single points of failure – including software.** Redundant hardware does not help if one faulty command or one automation component can affect every site at once. Limit the “blast radius” of changes.
2. **Validate changes before applying them.** Audit tools, staged roll-outs and automatic roll-back are essential; the audit tool itself must be tested.
3. **DNS is critical infrastructure.** Both outages were DNS-related (“it’s always DNS”). Use independent DNS providers, sensible TTLs and health-check logic that cannot take the whole service offline.
4. **Keep out-of-band management.** Management access must not depend on the network being fixed. Console servers, separate management networks and documented emergency procedures shorten recovery.
5. **Design for graceful degradation and multi-region failover.** Important services should keep working, possibly with reduced features, when one region or dependency fails.
6. **Monitor and practise.** Packet analysis (Wireshark), flow data and alerts help detect faults quickly, and regular disaster-recovery drills reveal weaknesses before real incidents do.

16. Relevance to the RNSB data centre

The measures we saw at RNSB – dual links and devices, firewalls, separate DMZ, UPS and diesel generator, a DR site and controlled physical access – directly address the failure modes of these two case studies. Redundant WAN links (Figure 2) protect against a single link failure; a tested DR site protects against loss of the primary site; and strict change control prevents a faulty maintenance command from spreading. The case studies also show that even very large organisations can fail, so documented procedures, out-of-band access and regular drills remain necessary for a bank.

17. Conclusion

The Facebook and AWS outages show that modern outages are rarely caused by a single broken cable; they arise from configuration errors, hidden software defects and dependencies between systems such as DNS and routing. Good network design combines redundancy, careful change management, independent management access, monitoring and regular testing. Studying these incidents deepened my understanding of BGP, DNS, redundancy and troubleshooting tools, and showed why the practices observed during the data-centre visit are important.

18. Appendix: Troubleshooting toolkit for outages

When an outage occurs, engineers work through the layers step by step. The table below lists the tools and commands I practised in Cisco Packet Tracer and Wireshark and what each one tells us.

Tool / command	What it checks	Example finding
ping	Layer 3 reachability and latency	Request timed out: path or device is down
tracert / traceroute	Hop-by-hop path to the destination	Packets stop at the hop where the link fails
show ip route / show ip bgp	Routing table and BGP routes (Packet Tracer / Cisco CLI)	Missing prefix means route was withdrawn
show interfaces / show ip interface brief	Interface status, errors, up/down state	Interface down / line protocol down
nslookup / dig	DNS name resolution	No answer or SERVFAIL for the service name
Wireshark filters (dns, bgp, icmp)	Packet-level evidence of the fault	Empty DNS answers, BGP withdrawals, retransmissions

Applying these steps in order – physical link, IP reachability, routing, DNS, application – quickly narrows an outage to its layer, which is exactly how the two incidents above were diagnosed.

References

1. Meta Engineering, “More details about the October 4 outage”, 5 October 2021 – engineering.fb.com/2021/10/05/networking-traffic/outage-details/
2. Amazon Web Services, “Summary of the Amazon DynamoDB Service Disruption in the Northern Virginia (US-EAST-1) Region”, October 2025 – aws.amazon.com/message/101925/
3. InfoQ, “Race Condition in DynamoDB DNS System: Analyzing the AWS US-EAST-1 Outage”, November 2025.
4. Rajkot Nagarik Sahakari Bank Ltd. – About Bank, Milestones and Our Strengths pages – rnsb.bank.in

Note: Figures 1–4 are illustrative diagrams drawn in the style of Cisco Packet Tracer and Wireshark to explain the concepts; they are not screenshots of live captures.